Phase Shifter

A dedicated phaser module for the FORMANT is the icing on the cake, musically enhancing smaller FORMANT versions (e.g., the “Mini-FORMANT”) in particular.

While the FORMANT VCO offers a distinctive feature compared to many commercial units—namely, pulse width modulation (PWM) for the square wave, which allows a phasing effect to be created using just a single VCO—this effect does not apply to the other waveforms. It is precisely in this context that a dedicated phase shifter module has proven advantageous.

Unlike high-pass and low-pass filters, all-pass filters maintain a constant output signal amplitude across a wide frequency range. Instead, they introduce a frequency-dependent phase shift between the input and output signals. This effect can be utilized to delay analog signals; consequently, such phase shifters are frequently employed in music electronics to produce the characteristic phasing effect.

Figure 1 illustrates how easily an all-pass filter can be implemented. In this circuit, the phase shift between the input and output signals depends solely on the values of resistor R and capacitor C. The diagram shows the phase shift as a function of frequency. It is evident from this that the circuit is a first-order all-pass filter, as the phase shift ranges from 0° to 180°.

Figure 2 shows the phaser circuit. IC1a ensures high input impedance. Its output drives a six-stage phase-shift chain constructed using operational amplifiers IC1b through IC2c. Field-effect transistors T1 through T6 operate as voltage-controlled resistors. The higher the drain-source resistance, the smaller the phase shift. Operational amplifier IC2d sums the phase-shifted and non-phase-shifted input signals, while potentiometer P2 allows the mini-phaser's output amplitude to be matched to various amplifier sensitivities. To produce a phasing effect, a triangle-wave generator is also required to drive the gates of FETs T1 through T6; this is built using operational amplifiers IC3 and IC4. The first operational amplifier functions as a non-inverting Schmitt trigger, with its hysteresis determined primarily by the positive feedback provided by resistors R26 and R27. It receives its input signal from an integrator consisting of the frequency-determining components R28, P1, C9, and IC4. The amplitude of the triangle-wave voltage at the integrator output ranges from -6 V to -4 V, as the FET gates must be negatively biased relative to the source. The triangle-wave generator can be disabled, allowing for manual phasing via potentiometer P3; this is particularly recommended for drums, as it enables the phasing effect to be optimally matched to the drummer's tempo. Potentiometer P4 controls the phasing depth—that is, the balance between the phase-shifted signal and the original signal. The phasing effect is most pronounced when the triangle-wave generator oscillates at a frequency of approximately 0.5 Hz to 1 Hz. If the frequency of this generator is increased to approximately 4 Hz, the phasing effect is lost, but a new one emerges: phase vibrato.

To compensate for the tolerances of the field-effect transistors, potentiometer P5 is provided.

The Front Panel

Figure 4 shows the front panel design for the phase shifter.

A single-pole miniature toggle switch suffices for the "automatic/manual frequency modulation" function (S2). The mounting holes for the potentiometers are designed for types with 6 mm shafts (P1 = automatic frequency modulation, P3 = manual frequency modulation, P4 = rate).

Switch S1 (on/off) allows the phasing effect to be switched off without having to unplug patch cords. Alternatively, this switch can be used to preset the desired phasing depth.

The Circuit Board

The circuit board layout (Figure 6) has been redesigned specifically for use in 19" rack-mount enclosures and card magazines. The additional wiring for the power supply lines is shown in Figure 5. A 31-pin connector strip is used. Otherwise, the component placement diagram is identical to that shown in Figure 3. 4

BOM

Resistors:

R1, R2 = 100k
R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R21, R22 = 22k
R15, R16, R17, R18, R19, R20, R23, R35 = 10k
R24, R25 = 33k
R26 = 330k
R27 = 1M
R28, R36 = 4k7
R29 = 10M
R30 = 12k
R31 = 3k9
R32, R34 = 47k
R33 = 470k

P1 = 470 K lin.
P2,P5 = БОК trimmer
P3 = 10k lin.
P4 = 10k log.

Capacitors:

C1 = 470n
C2 = 4µ7/30 V tantalum
C3 = 1n
C4 = 470n
C5 = 330n
C6 = 150n
C7 = 100n
C8 = 47n
C9 = 4µ7/16V tantalum

Semiconductor:

IC1, IC2 = 324 (better XR 4212)
IC3, IC4 = 741
T1, T2, T3, T4, T5, T6 = BF 245C

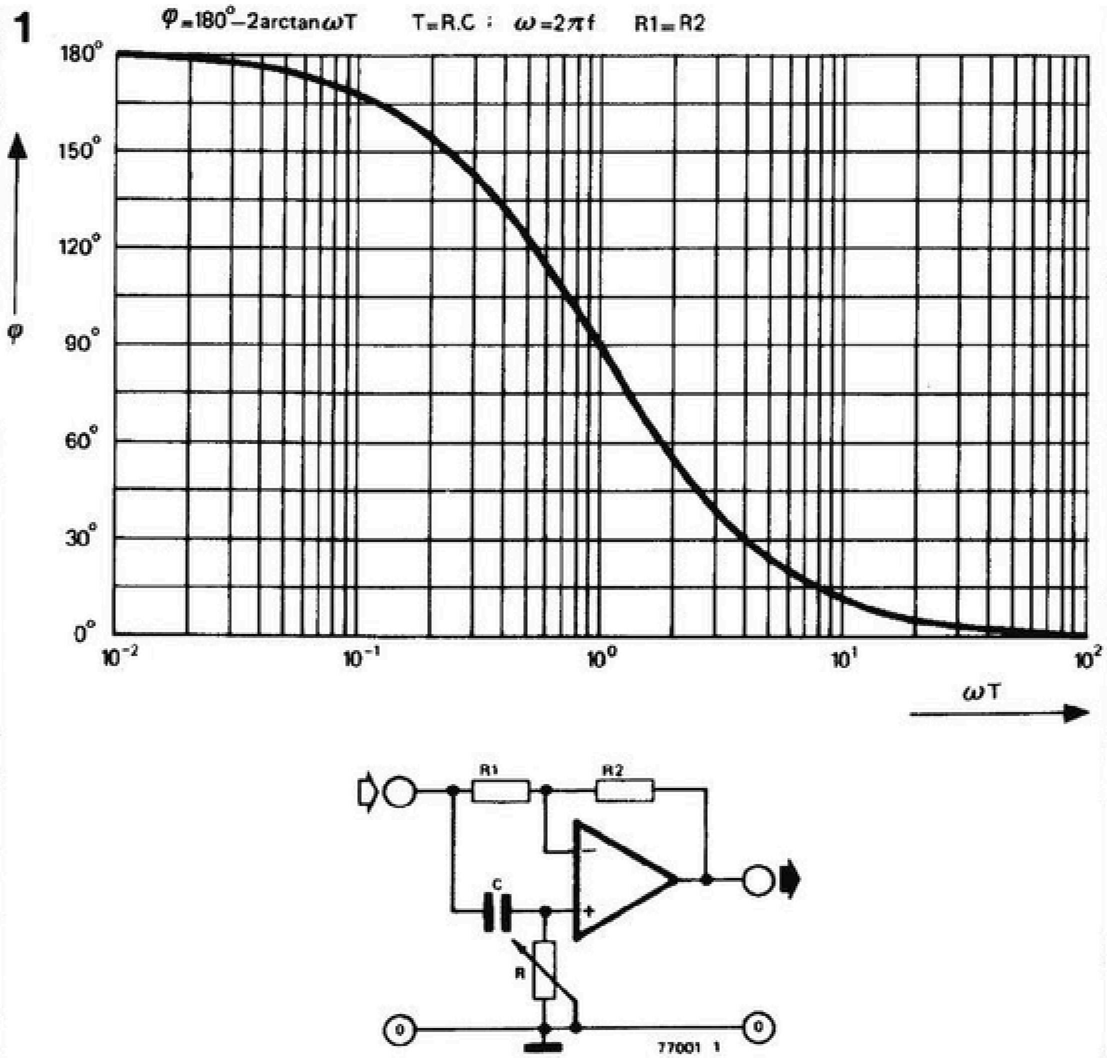
Miscellaneous:

S1, S2 = Switch 1 x Um

Figures

- Figure 1. Basic circuit configuration of a high-pass filter. The diagram shows the phase shift as a function of frequency.
- Figure 2. Complete phaser circuit.
- Figure 3. PCB layout and component placement diagram for the circuit shown in Figure 2.
- Figure 4. Proposed front panel design for the phase shifter.
- Figure 5. The supply voltage for the phaser is +12 V. The required voltage can be derived from the +15 V FORMANT supply voltage using just a few components.
- Figure 6. PCB layout and component placement diagram for the circuit shown in Figure 2. The PCB is in Eurocard format and includes the additional circuitry from Figure 5.

1



φ = 180° − 2arctanωT   T = R.C ; ω = 2πf   R1 = R2
180°
150°
120°
90°
60°
30°
0°
φ
10⁻²
10⁻¹
10⁰
10¹
10²
ωT
R1
R2
C
R
0
0
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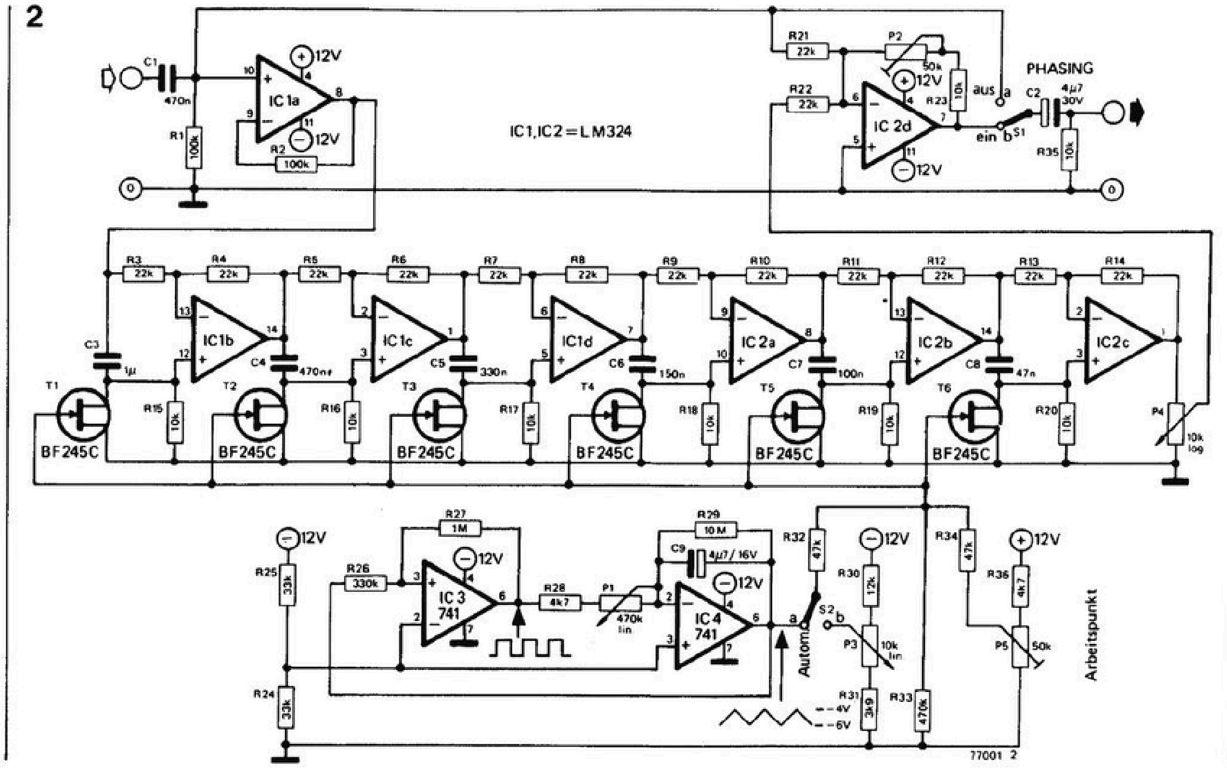
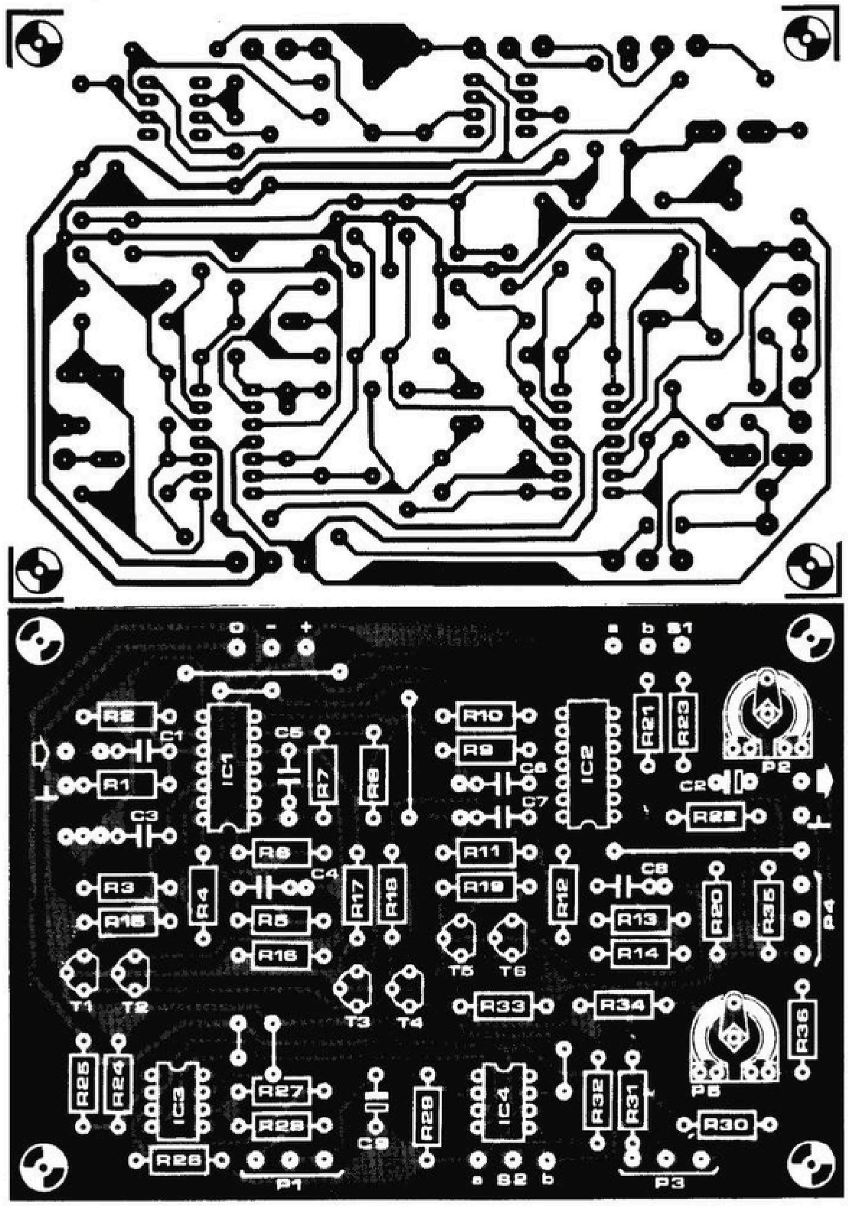
2



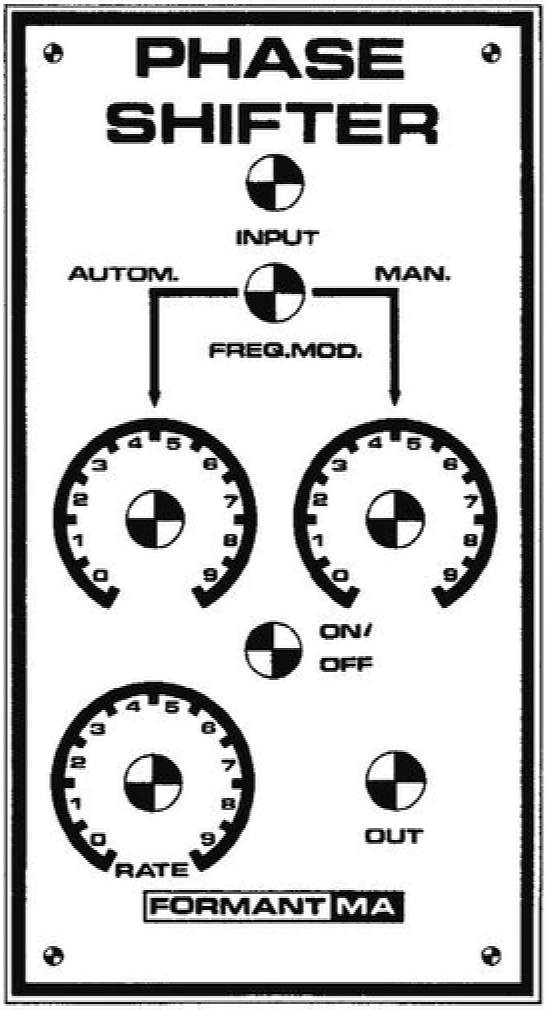
C1
470n
R1
100k
IC 1a
R2
100k
12V
IC1, IC2 = LM324
R21
22k
P2
50k
R22
22k
IC 2d
R23
10k
aus
a
ein
b
S1
PHASING
C2
4μ7
30V
R35
10k
R3
22k
R4
22k
R5
22k
R6
22k
R7
22k
R8
22k
R9
22k
R10
22k
R11
22k
R12
22k
R13
22k
R14
22k
IC1b
IC1c
IC1d
IC2a
IC2b
IC2c
C3
1μ
C4
470n
C5
330n
C6
150n
C7
100n
C8
47n
T1
T2
T3
T4
T5
T6
BF245C
R15
10k
R16
10k
R17
10k
R18
10k
R19
10k
R20
10k
P4
10k
log
R25
33k
R24
33k
R26
330k
R27
1M
IC 3
741
R28
4k7
P1
470k
lin
R29
10M
C9
4μ7 / 16V
IC 4
741
R32
47k
S2
Autom
R30
12k
P3
10k
lin
R31
3k9
R33
470k
-4V
-6V
R34
47k
R36
4k7
P5
50k
Arbeitspunkt
77001 2

3

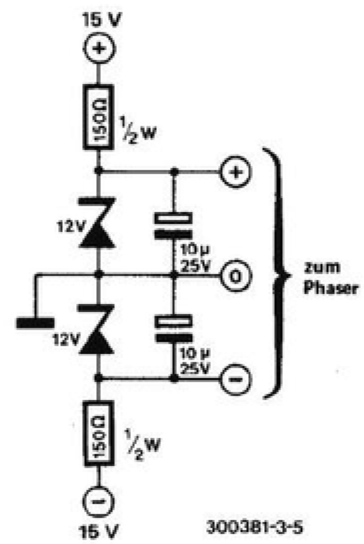


0 - +
a b S1
R2
C1
R1
C3
IC1
C5
R7
R8
R6
C4
R4
R5
R16
R17
R18
R3
R15
T1 T2
T3 T4
R10
R9
C6
C7
IC2
R21
R23
P2
C2
R22
R11
R19
R12
C8
R13
R14
R20
R35
P4
T5 T6
R33
R34
R36
P5
R25
R24
IC3
R27
R28
R26
P1
C9
R29
IC4
a S2 b
R32
R31
R30
P3

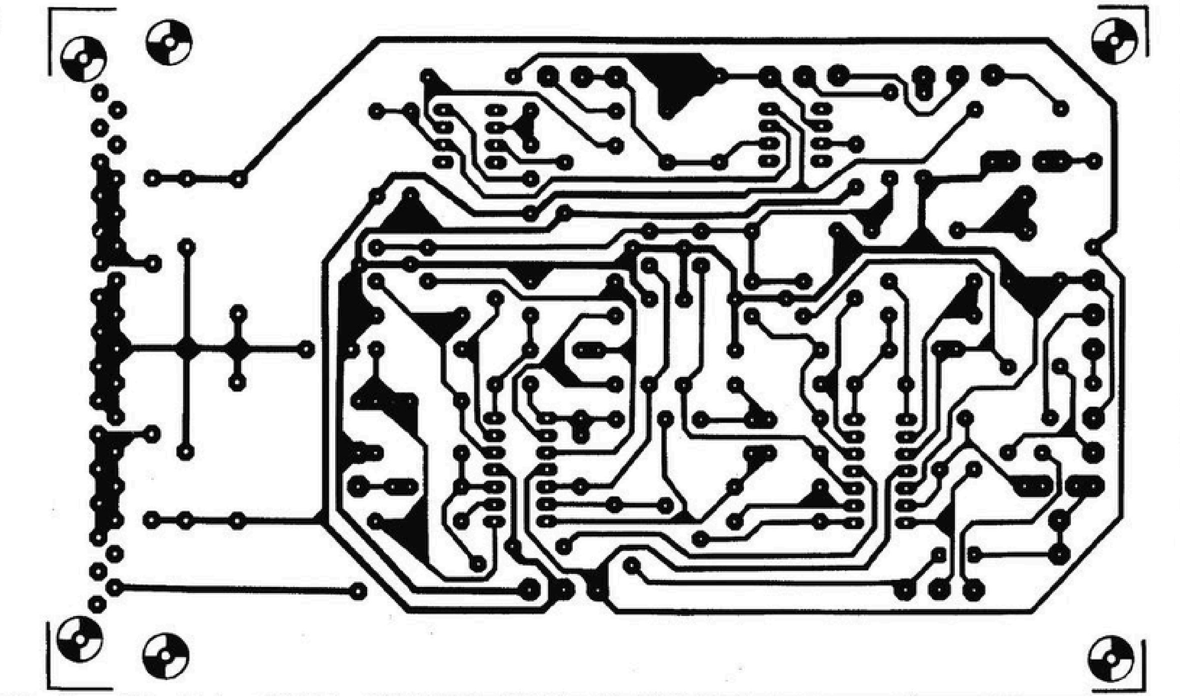
4

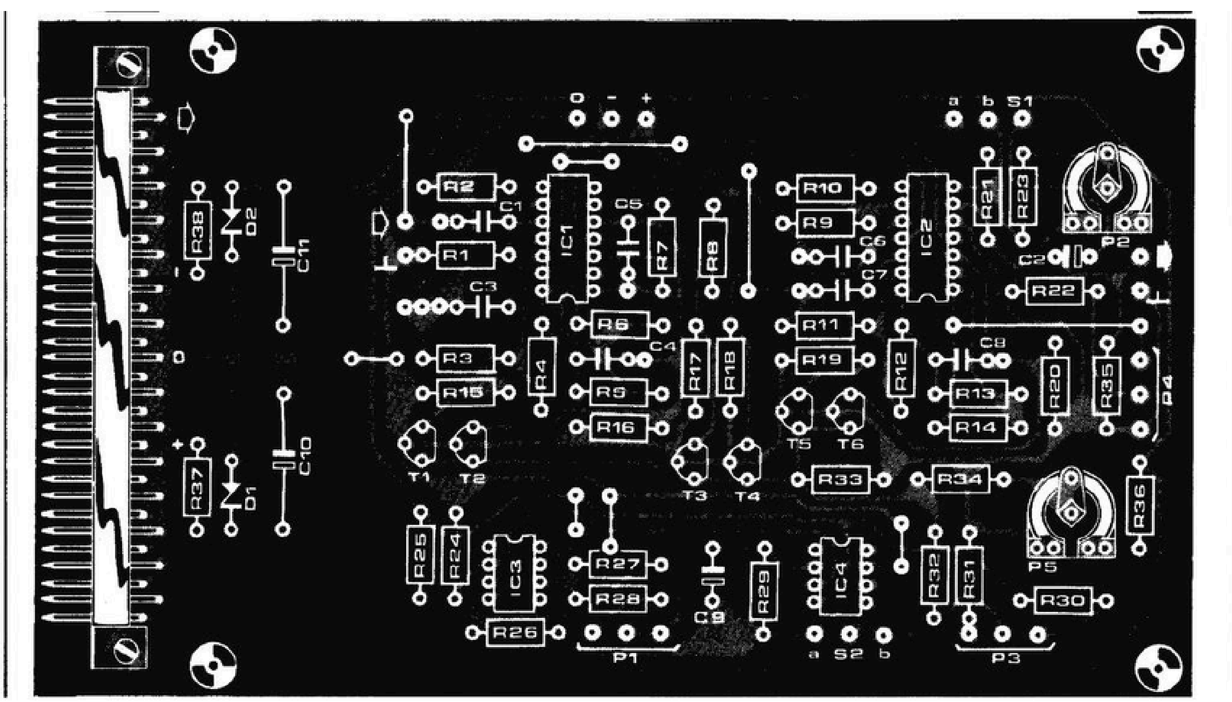

PHASE
SHIFTER
INPUT
AUTOM.
MAN.
FREQ.MOD.
ON/
OFF
RATE
OUT
FORMANT MA

5


15 V
150Ω
1/2 W
12V
10 µ
25V
zum
Phaser
15 V
300381-3-5

6




0 - +
a b S1
R2
C1
C5
R7
R8
IC1
R10
R9
C6
C7
IC2
R21
R23
P2
C2
R22
R1
C3
R6
C4
R11
R19
R12
C8
R20
R35
P4
R3
R4
R17
R18
R15
R5
R16
T5
T6
R13
R14
T1
T2
T3
T4
R33
R34
R36
P5
R38
D2
C11
R37
D1
C10
R25
R24
IC3
R27
R28
C9
R29
IC4
R32
R31
R30
R26
P1
a S2 b
P3